

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 1

Kopiereg voorbehou

Module EBN111
Elektrisiteit en Elektronika
12 Maart 2015

Semester Test 1

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Module EBN111
Electricity and Electronics
12 March 2015



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute <i>90 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		17	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert
	2 A Chatterjee

VRAAG/QUESTION 1 [22]

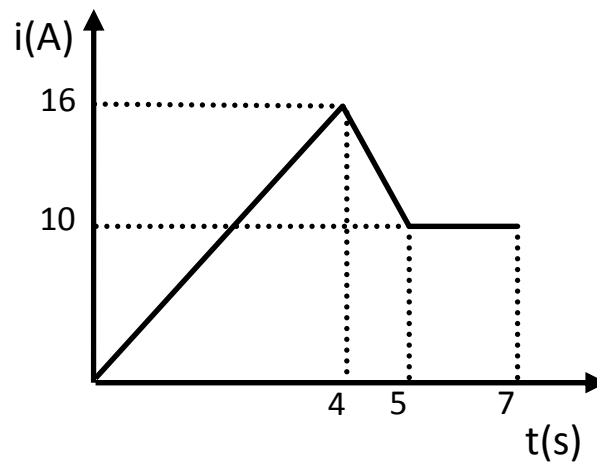
Vraag 1.1 [3] a) Hoeveel elektrone word verteenwoordig deur 2C lading? b) 'n Stoof met twee plate, elk gespesifiseer as 1.5kW, word gebruik om 'n maaltyd voor te berei. Die gebruik van elke plaat is as volg: Plaat 1: 15 minute, Plaat 2: 20 minute. As energie se koste 50 sent/kWh is, bereken die koste (in Rand) om die maaltyd voor te berei.	Question 1.1 [3] a) How many electrons are represented by 2C of charge? b) A stove with 2 hotplates, rated at 1.5kW each, is used to prepare a meal. The duration of its usage is described as follows: Plate 1: 15 minutes, Plate 2: 20 minutes. If the energy charge is 50 cents/kWh, calculate the total cost (in Rands) of preparing the meal.
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Vraag 1.2 [2]

Die stroom deur 'n element word hieronder getoon.
Hoeveel lading het deur die element beweeg teen $t = 5\text{s}$.

Question 1.2 [2]

The current through an element is shown below. What is the total charge that passed through the element at $t = 5\text{s}$.



Vraag 1.3 [2]

Die stroom wat by die positiewe terminaal van 'n toestel ingaan is $i(t) = 5t^2 + 2$ A. Die spanning oor die toestel is $v(t) = 4 \frac{di}{dt}$ V. Bepaal die drywing ge-assosieer met die toestel by $t = 1$ s.

Question 1.3 [2]

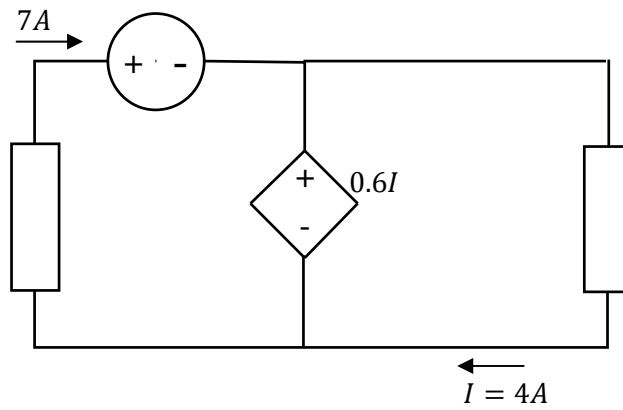
The current entering the positive terminal of a device is $i(t) = 5t^2 + 2$ A. The voltage across the device is $v(t) = 4 \frac{di}{dt}$ V. Find the power associated with the device at $t = 1$ s.

Vraag 1.4(a) [2]

Bepaal die drywing ge-assosieer met die afhanklike bron.

Question 1.4(a) [2]

Find the power associated with the dependent source.

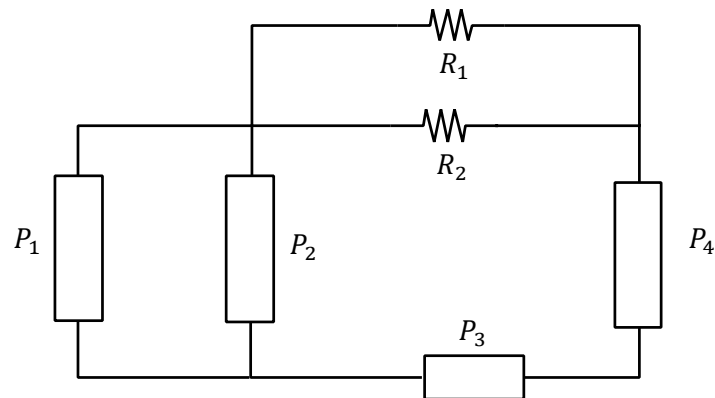


Vraag 1.4(b) [2]

Die figuur bevat 'n stroombaan met 6 komponente. Bepaal die drywing ge-assosieer met R_1 as $P_1 = -200\text{W}$, $P_2 = 55\text{W}$, $P_3 = 40\text{W}$, $P_4 = 45\text{W}$. R_1 en R_2 het dieselfde weerstandswaarde.

Question 1.4(b) [2]

The figure below shows a circuit with 6 elements. Find the power associated with R_1 if $P_1 = -200\text{W}$, $P_2 = 55\text{W}$, $P_3 = 40\text{W}$, $P_4 = 45\text{W}$. R_1 and R_2 have the same resistance value.

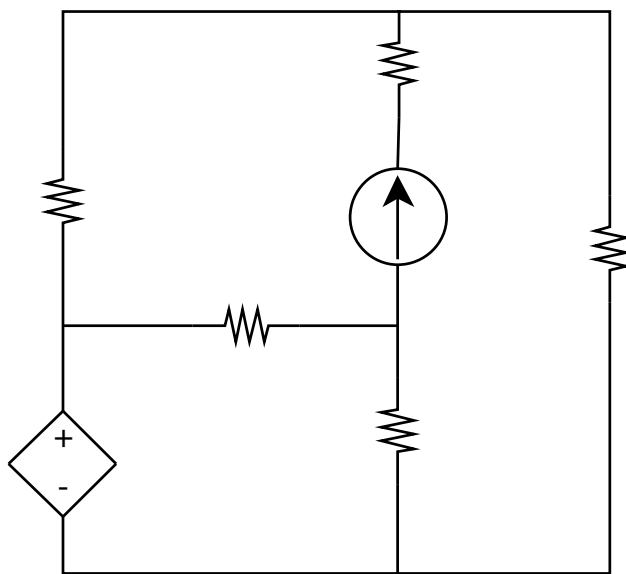


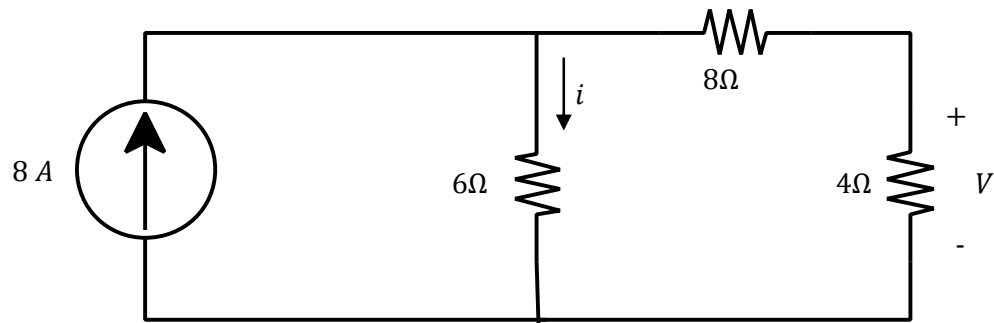
Vraag 1.5 [3]

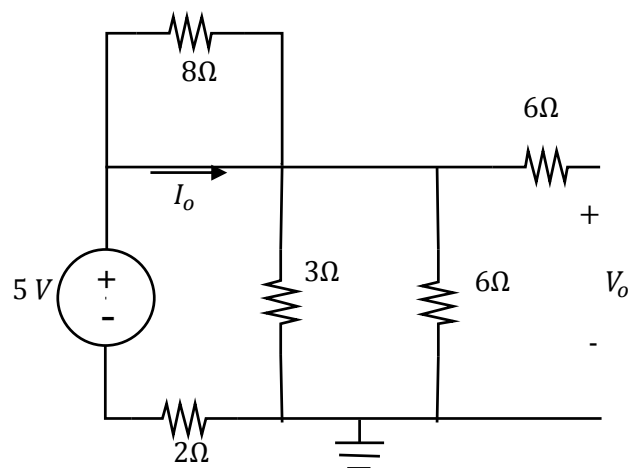
Bereken die aantal nodes, takke, en onafhanklike lusse in die stroombaan (n , b en l).

Question 1.5 [3]

Determine the number of nodes, branches, and independent loops in the circuit below (n , b and l).



Vraag 1.6 [2]Bepaal i en V in in die stroombaan.**Question 1.6 [2]**Find i and V in the circuit below.

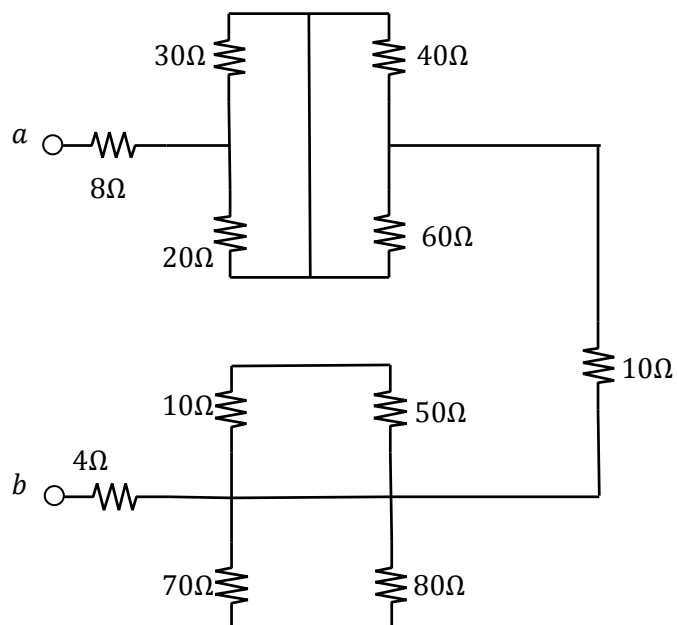
Vraag 1.7 [2]Bepaal I_o en V_o in die stroombaan.**Question 1.7 [2]**Find I_o and V_o in the circuit below.

Vraag 1.8(a) [2]

Bepaal die ekwivalente weerstand tov terminale a-b.

Question 1.8(a) [2]

Find the equivalent resistance wrt terminals a-b.

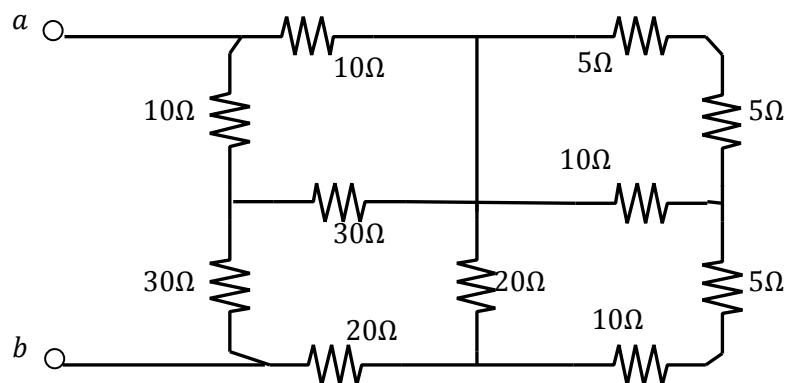


Vraag 1.8(b) [2]

Bepaal die ekwivalente weerstand tov terminale a-b.

Question 1.8(b) [2]

Find the equivalent resistance wrt terminals a-b.



VRAAG/QUESTION 2 [23]

Vraag 2.1 [4]

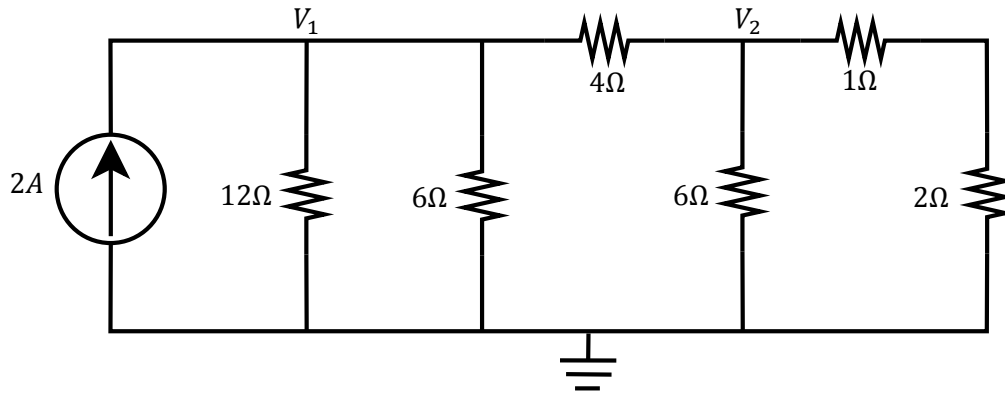
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

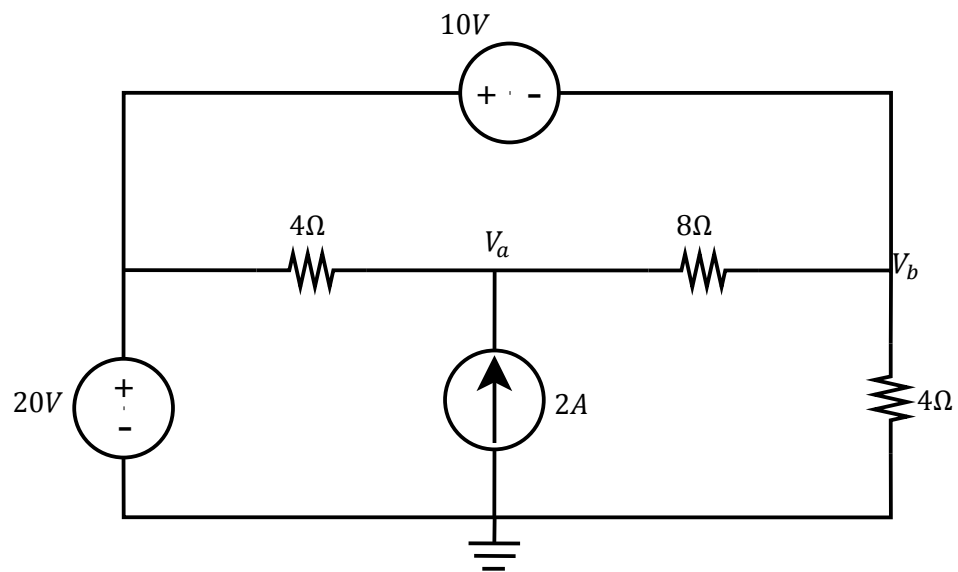
$$\begin{aligned} aV_1 + bV_2 &= 48 \\ -3V_1 + cV_2 &= d \end{aligned}$$

Question 2.1 [4]

Use nodal analysis to set up two equations of the following form:

$$\begin{aligned} aV_1 + bV_2 &= 48 \\ -3V_1 + cV_2 &= d \end{aligned}$$



Vraag 2.2 [4]Bereken V_a en V_b .**Question 2.2 [4]**Determine V_a and V_b .

Vraag 2.3 [2]

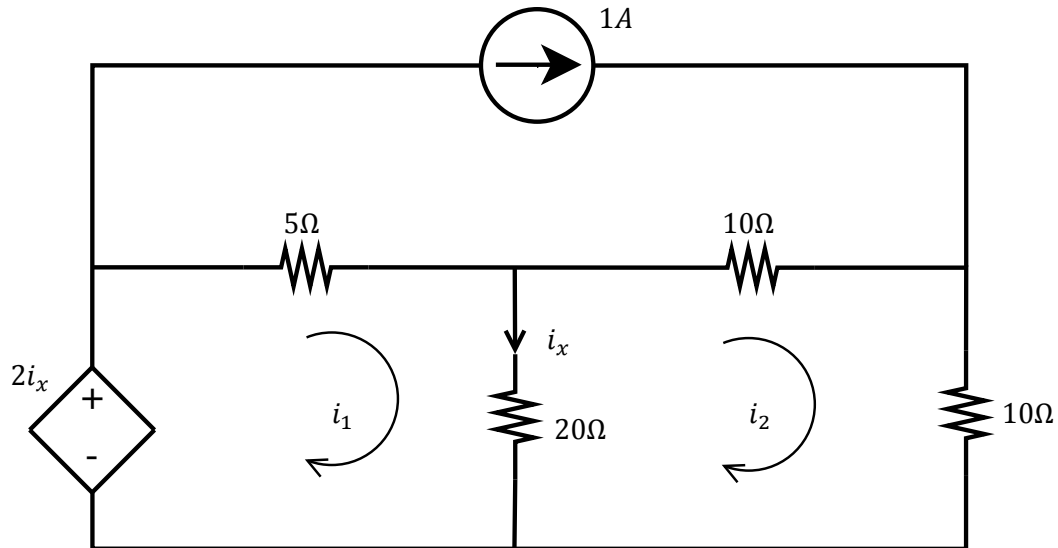
Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel:

$$23i_1 + ei_2 = f$$
$$gi_1 + hi_2 = 10$$

Question 2.3 [2]

Use mesh analysis to set up two equations of the following form:

$$23i_1 + ei_2 = f$$
$$gi_1 + hi_2 = 10$$

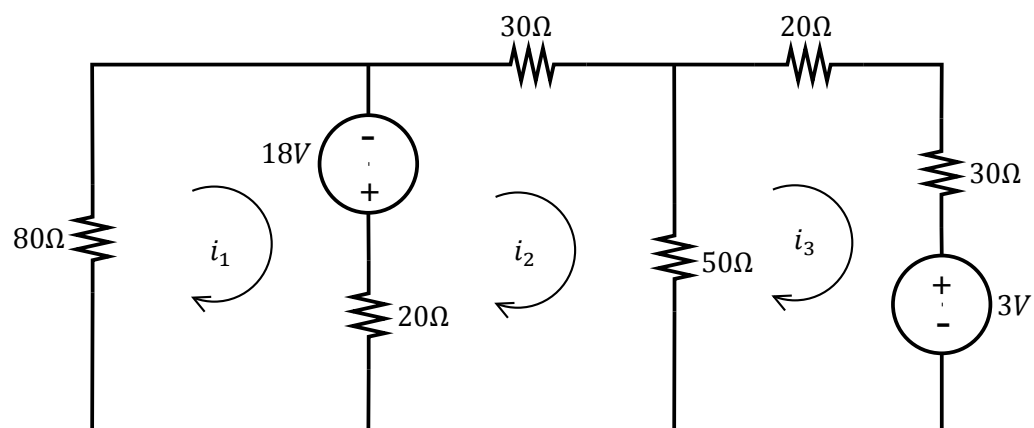


Vraag 2.4 [3]

Stel 'n matriksvergelyking d.m.v. inspeksie op in terme van die lusstrome.

Question 2.4 [3]

Use inspection to set up a matrix equation in terms of the mesh currents.



Vraag 2.5 [4]a) Bepaal i_2 .

$$\begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -10 \end{bmatrix}$$

b) Bepaal V_3 .

$$\begin{bmatrix} 5 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix}$$

Question 2.5 [4]a) Determine i_2 .

$$\begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -10 \end{bmatrix}$$

b) Determine V_3 .

$$\begin{bmatrix} 5 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix}$$

Vraag 2.6 [4]

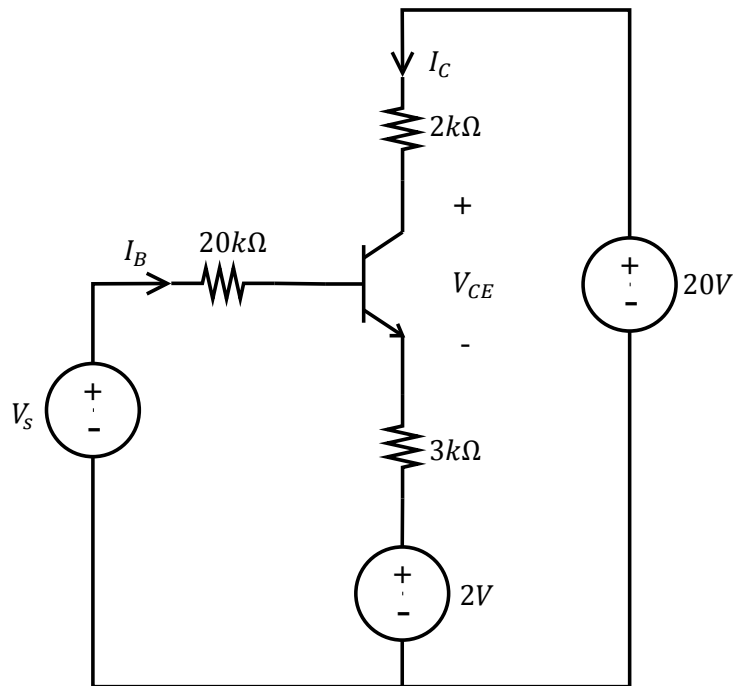
Gegee: $\beta = 100$; $V_{BE} = 0.7 \text{ V}$.

- a) Indien $V_S = 10 \text{ V}$, bepaal I_B (in μA).
b) Indien $V_{CE} = 10 \text{ V}$, bepaal I_C (in mA).

Question 2.6 [4]

Given: $\beta = 100$; $V_{BE} = 0.7 \text{ V}$.

- a) If $V_S = 10 \text{ V}$, determine I_B (in μA).
b) If $V_{CE} = 10 \text{ V}$, determine I_C (in mA).



EBN111 SEMESTER TOETS/TEST 1 (12 MAART/MARCH 2015)
ANTWOORDBLAD / ANSWER SHEET

Van / Surname:		Studentenommer: Student number:							
Voorletters / Initials:		Taal / Language:			Afr		Eng		

Q1		Q2		Tot:	/45
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VRAAG 1 – QUESTION 1 (22)			
1.1	$a) n =$	$b) Cost =$	
1.2	$Q =$		
1.3	$P =$		
1.4	$a) P_{(0.6I)} =$	$b) P_{(RI)} =$	
1.5	$n =$	$b =$	$l =$
1.6	$i =$	$V =$	
1.7	$I_o =$	$V_o =$	
1.8	$a) R_{eq} =$	$b) R_{eq} =$	

VRAAG 2 – QUESTION 2 (23)				
2.1	$a =$	$b =$	$c =$	$d =$
2.2	$V_a =$		$V_b =$	
2.3	$e =$	$f =$	$g =$	$h =$
2.4	$\begin{bmatrix} & & & & \\ & & & & \\ & & & & \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} \\ \\ \end{bmatrix}$			
2.5	$i_2 =$		$V_3 =$	
2.6	$I_C =$		$I_B =$	